5.3.3 Cast wheel

The modern automotive wheels are generally made up of aluminum or magnesium alloy materials. Figure 5.3.3 shows the photographic view of a cast wheel. Cast wheels are mostly used in cars. The simple structure of this wheel can be seen in figure. Forged wheels are mostly used in heavy vehicles. These types of wheels are less in weight. The wheels made of magnesium alloy are 50% less than the weight of the steel wheel, and the aluminum alloy wheels are 70% less than the weight of the steel wheel. Even though the weight is lower than that of steel wheels these wheels have the strength which is equal to the steel wheels. These metal alloys have the ability to transfer heat and reduce the heat generated in the tyres or brakes. Moreover these materials are used for manufacture the wide rims. This gives more vehicle stability. Magnesium alloy material has the ability to withstand heavy loads. It could also withstand more vibrations and shocks occurred in vehicles.



Figure 5.3.3 A view of Cast Wheel

5.4 Wheel Dimension

The measurement of wheel dimensions could be seen in the Figure 5.4. The wheel dimension has the following letters represented as

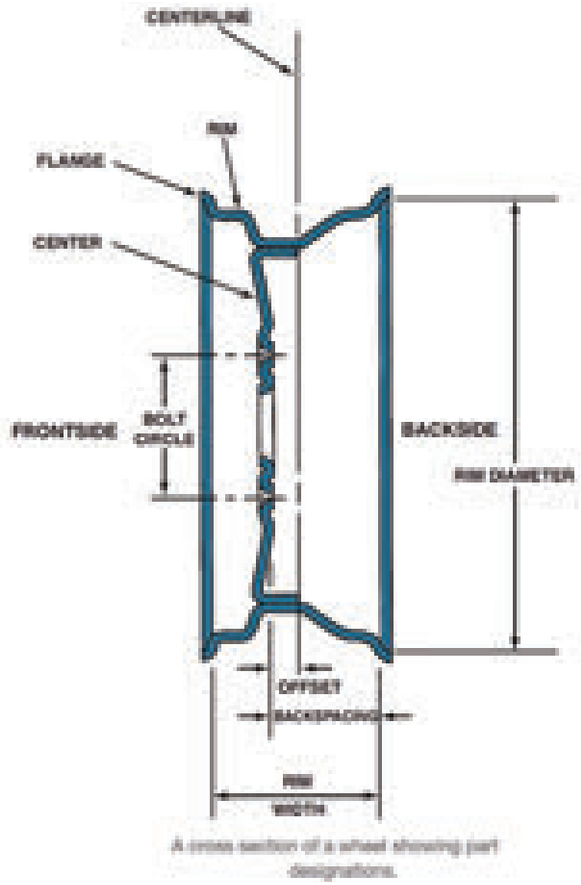


Figure 5.4 Wheel Dimensions

'W' - Width of the wheel.

'D' - Diameter of wheel.

'C' - Type of rim.

Wheels are shown by code number. For example, if the code number of the wheel is given as 5.50 B-13 and 4 ½ J-14 then, it indicates the terms as below.

Wheel Dimensions

Width (W)	Rim type (C)	Diameter (D)
5.50"	B	13"
4 ½"	J	14"

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